Topic 2: Making Your First API Calls (CRUD Operations)

API Integration Hands-On Course for n8n (Enterprise Level 3)

1. Document Control

Field	Specification Value
Document Title	Topic 2: Making Your First API Calls (CRUD Operations) Specification
Course Module	API Integration Hands-On for n8n (Level 3 - Enterprise Series)
Version	1.0 (Production Master)
Status	**APPROVED & VERIFIED**
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Prepared By	Lead Solutions Architect & Automation Engineering Team
Target Audience	Integration Engineers, API Developers, Automation Trainees

2. Learning Concepts & Architectural Principles

Representational State Transfer (REST) relies on standardized HTTP methods to perform operations on server resources:

- **GET (Read):** Retrieves resource state without side effects (Idempotent & Safe). Supports pagination, filtering, and sorting via URL Query Parameters (`?page=2`).
- **POST (Create):** Submits data in the request body to create a subordinate resource. Returns `201 Created` along with a server-generated identifier and timestamp.
- **PUT (Full Replace):** Overwrites an entire resource representation at a specific URI (Idempotent). Returns `200 OK` with update confirmation.
- **DELETE (Remove):** Purges the targeted resource from the server. Returns `204 No Content` or `200 OK` .

HTTP Status Code Classification

Understanding server response codes is critical for building resilient automation error handling:

- **`200 OK` :** Standard success for GET and PUT queries.
- **`201 Created` :** Successful resource instantiation via POST.
- **`204 No Content` :** Action succeeded without payload return (standard for DELETE).
- **`4xx Client Errors` :** Request contains invalid syntax, missing parameters, or nonexistent resource identifiers.

3. Business Requirements Document (BRD)

ID	Requirement	Business Rationale	Priority
BRD-001	Full CRUD Lifecycle Validation	Master all 4 primary HTTP verbs within a single unified workflow to eliminate custom scripts.	HIGH
BRD-002	Structured Query Parameterization	Support URL query string variables for pagination and resource filtering, avoiding string concatenation.	HIGH
BRD-003	JSON Body Serialization	Accurately construct and transmit JSON payloads in POST and PUT operations with correct headers.	CRITICAL
BRD-004	Status Code Dissection	Correctly interpret 200, 201, and 204 response semantics across tools to avoid false errors.	HIGH
BRD-005	Zero-Blocker PinData Coverage	Provide pre-pinned mock data across all nodes for offline execution.	MEDIUM

4. Product Requirements Document (PRD)

ID	Product Capability	User Value & Acceptance Criteria	Priority
PRD-001	Paginated GET Request	Fetches page 2 of users via query parameter (`page=2`); verifies 200 OK.	High
PRD-002	POST Resource Creation	Submits Alice Smith profile via JSON; receives 201 Created with generated ID and timestamp.	Critical
PRD-003	PUT Resource Replacement	Updates Alice Jones to Lead Engineer via PUT; confirms 200 OK and updatedAt.	High
PRD-004	DELETE Resource Removal	Issues DELETE to /api/users/2; verifies 204 No Content without error interruption.	High
PRD-005	Lifecycle Audit Summary	Synthesizes execution summary verifying 100% PASS across all 4 HTTP methods.	High

5. Software Requirements Specification (SRS) & Canvas Blueprint



6. Detailed Node Configuration Blueprint

Node Name	Type & Version	Operational Logic & Configuration
Manual_Trigger	manualTrigger (v1)	Canvas trigger initiating manual testing execution on button click.
HTTP_GetUsersPage2	httpRequest (v4.2)	Method: GET, URL: https://reqres.in/api/users, Query: page=2. Returns 6 user items.
HTTP_CreateUser	httpRequest (v4.2)	Method: POST, URL: https://reqres.in/api/users, JSON body creating Alice Smith.
HTTP_UpdateUser	httpRequest (v4.2)	Method: PUT, URL: https://reqres.in/api/users/2, JSON body updating Alice Jones.
HTTP_DeleteUser	httpRequest (v4.2)	Method: DELETE, URL: https://reqres.in/api/users/2, handles HTTP 204 empty response.
Code_AuditSummary	code (v2)	Consolidates end-to-end execution receipt verifying all 4 HTTP methods.

7. Testing & Quality Assurance Verification Plan

Scenario ID	Test Condition	Input Profile	Expected Result & Verification
TC-02-01	Query Parameter GET	page=2	Status 200 OK, returns 6 records starting with ID 7 (Michael Lawson).
TC-02-02	Payload POST Creation	Alice Smith JSON body	Status 201 Created, returns generated ID 721 and createdAt timestamp.
TC-02-03	Resource PUT Update	Alice Jones JSON body	Status 200 OK, returns updated title and updatedAt timestamp.
TC-02-04	Resource DELETE Removal	Target resource ID 2	Status 204 No Content, resource purged from mock server.
TC-02-05	Complete Canvas Run	1-Click Execute workflow	100% green checkmarks across all canvas nodes.